Chapter 6

From Periodontal Tactile Function to Peri-implant Osseoperception

Reinhilde Jacobs

Laboratory of Oral Physiology, Department of Periodontology, and Oral Imaging Center, Faculty of Medicine, Catholic University of Leuven, Leuven, Belgium

Introduction

Perception is the ability to detect external stimuli through vision, audition, balance, somatic function, taste or smell (Martin 1991). In this chapter, only perception via the somatosensory system is explored. The preponderance of this system is illustrated by its major representation, along with that of the hand, in the sensory homunculus proposed by Penfield (Penfield & Rasmussen 1950). Generally speaking, the somatosensory function is essential for the fine-tuning of limb movements.

Likewise, tactile function of teeth plays a crucial role in refinement of jaw motor control. Periodontal mechanoreceptors and especially those located in the periodontal ligament, are extremely sensitive to external mechanical stimuli (Jacobs & van Steenberghe 1994). The periodontal ligament can thus be considered as a keystone for masticatory and other oral motor behaviors. Any condition that may influence periodontal mechanoreceptors could alter the sensory feedback pathway and thus affect tactile function and fine-tuning of jaw motor control (e.g. periodontal breakdown, bruxism, re-implantation, anesthesia).

The most dramatic change may occur after extraction of teeth, as this eliminates all periodontal ligament receptors. This condition may persist after implant placement, as functional re-innervation has not yet been proven in humans. Surprisingly, patients with implant-supported prostheses often seem to function quite well. The underlying mechanism of this so-called "osseoperception" phenomenon remains a matter of debate, but the response of assumed peri-implant receptors might help to restore the proper peripheral feedback pathway. This hypothesized physiologic integration might thus lead to better acceptance, improved psychological integration, and more natural functioning.

This chapter will unravel periodontal tactile function and guide the reader through the mysteries of peri-implant osseoperception in order to find neuroanatomic, histologic, physiologic, and psychophysical evidence to confirm the hypothesis.

Clinical Periodontology and Implant Dentistry, Sixth Edition. Edited by Niklaus P. Lang and Jan Lindhe.

Neurophysiologic background

Trigeminal neurosensory pathway

The sensory inputs of the oral region are carried by the mandibular and maxillary divisions of the trigeminal nerve via the trigeminal ganglion to the brainstem. This is part of an important sensory feedback pathway that refines jaw movements. The afferent signals are transmitted either to the main sensory nucleus of the trigeminal nerve (responsive to tactile senses, light touch, and pressure) or to the descending spinal tract nuclei (responsive to sensation of oral mucosa, pain, temperature, and deep pressure). From these, signals are transferred across the midline and sent to the thalamus and, via thalamocortical projections, to the respective cortical areas involved in orofacial sensation, where they can result in conscious perception.

Neurovascularization of the jaw bones

The jaws are richly supplied by neurovascular structures, and it is thus of utmost importance to identify vital anatomic structures before carrying out a surgical procedure. During a radiographic preoperative planning procedure, neurovascular structures need to be precisely localized so that they can be avoided in surgical treatment. Studies reveal that edentulous and dentate anterior jaws present significant variation in jaw bone neurovascularization (for review see Jacobs *et al*. 2007). All these canal structures contain a neurovascular bundle whose diameter may be large enough to cause clinically significant trauma if damaged. Surgeons need to avoid the nervous structures to avoid this trauma, and also because these critical structures may be essential in the potential re-innervation of peri-implant bone. Indeed, the remaining neurovascular bundles in the edentulous jaw bone may support nerves that regenerate after tooth extraction and implant placement. This assumption is the basis of the so-called osseoperception phenomenon and will be further outlined below.

Mandibular neuroanatomy

The mandibular nerve runs forward through the mandibular canal. At the mental foramen it gives off the mental nerve. Additional mental foramina exist, with a reported prevalence of 9%. These canals are often smaller and located more posteriorly (Oliveira-Santos *et al*. 2011). Meanwhile, the mandibular nerve continues to run anteriorly as the mandibular incisive nerve (Mraiwa *et al*. 2003a, b) (Fig. 6-1). Conventional intraoral and panoramic radiographs often fail to show the so-called incisive canal (Jacobs *et al*. 2004). Cross-sectional imaging may, however, be used to locate the canal and as such protect it from the risk for neurovascular damage (Jacobs *et al*. 2002a) The mandibular incisive canal undeniably contains a true neurovascular bundle with nervous sensory structures (Fig. 6-2). Its existence in edentulous patients is underlined by reported surgical complications. Indeed, sensory disturbances, caused by direct trauma to the mandibular incisive canal bundle, have been reported after implant placement in the interforaminal region (Jacobs *et al*. 2007) (Fig. 6-3). Sensory disorders may also result from indirect trauma caused by a hematoma in the canal, and as the latter acts as a closed chamber, pressure is exerted on the mandibular incisive canal bundle and this spreads to the main mental branch (Mraiwa *et al*. 2003b).

Other anatomic landmarks to be noted are the superior and inferior genial spinal foramina and their bony canals, situated in the midline of the mandible in 85–99% of mandibles (Liang *et al*. 2005a, b; Jacobs *et al*. 2007) (Fig. 6-4). The superior genial spinal foramen has been found to contain a branch of the lingual artery, vein, and nerve. Furthermore, a branch of the mylohyoid nerve together with branches or anastomoses of sublingual and/or submental arteries and veins have been identified in the inferior genial spinal foramen. This artery can be of a size that, if it is damaged, a hemorrhage intraosseously or in the connective soft tissue can be provoked and may be difficult to control (Liang *et al*. 2005a, b) (Figs. 6-5, 6-6).

Fig. 6-1 These human dry mandibular bone sections illustrate the presence and dimensional importance of the mandibular incisive nerve, even in edentulism. The middle section actually shows the mandibular midline, confirming that there is no true connection between the left and right sections.

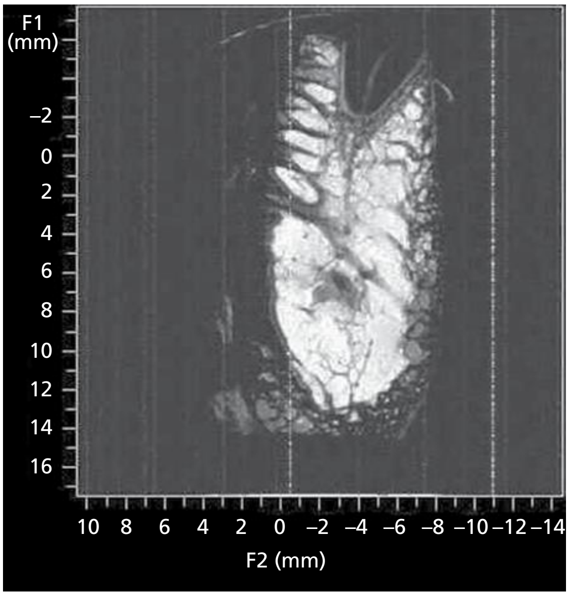


Fig. 6-2 Single cross-sectional slice of a high resolution MRI dataset localized at the incisor region of a dentate anterior human mandible, with the fatty marrow colored white.
A black root-form structure corresponds to the root of an incisor tooth. It is surrounded by a small band of intermediate signal intensity, representing the periodontal ligament. Furthermore, the dental neurovascular supply is seen as a line of intermediate signal intensity in the middle of the root. The latter is descending to the level of a larger structure of intermediate signal intensity (incisive nerve) with a black oval area on top (vascular structure). (Source: Jacobs *et al*. 2007. Reproduced with permission from Elsevier.)

Fig. 6-3 Cross-sectional slice of a cone-beam CT dataset showing an osseointegrated implant placed in an edentulous lateral incisor region, on top of a prominent incisive canal lumen. Chronic pressure on the incisive nerve resulted in neuropathic pain. (Source: Jacobs & Steenberghe 2006. Reproduced with permission from John Wiley & Sons.)

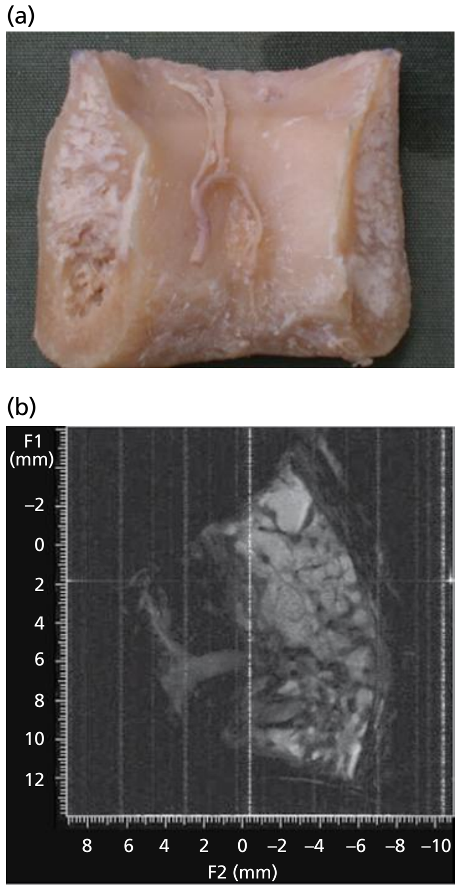


Fig. 6-4 (a) Macroanatomic view of a human anterior mandible showing a clear neurovascular bundle entering the superior genial spinal foramen. (b) Matching horizontal slice acquired through high-resolution MRI confirms the entry of a neurovascular bundle into the superior genial spinal foramen. (Courtesy of I. Lambrichts, University of Hasselt.)

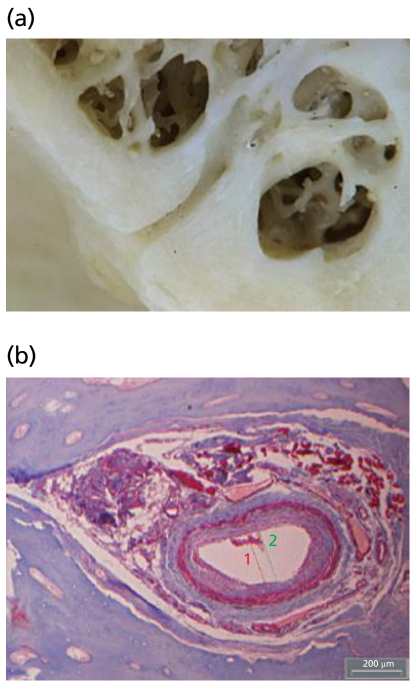


Fig. 6-5 (a) Stereomicroscopic image showing a single genial spinal canal section. (b) Neurovascular content of the canal is confirmed histologically. The artery in this image has a diameter of about 0.5 mm (red and green lines denote the inner and outer wall dimensions, respectively).

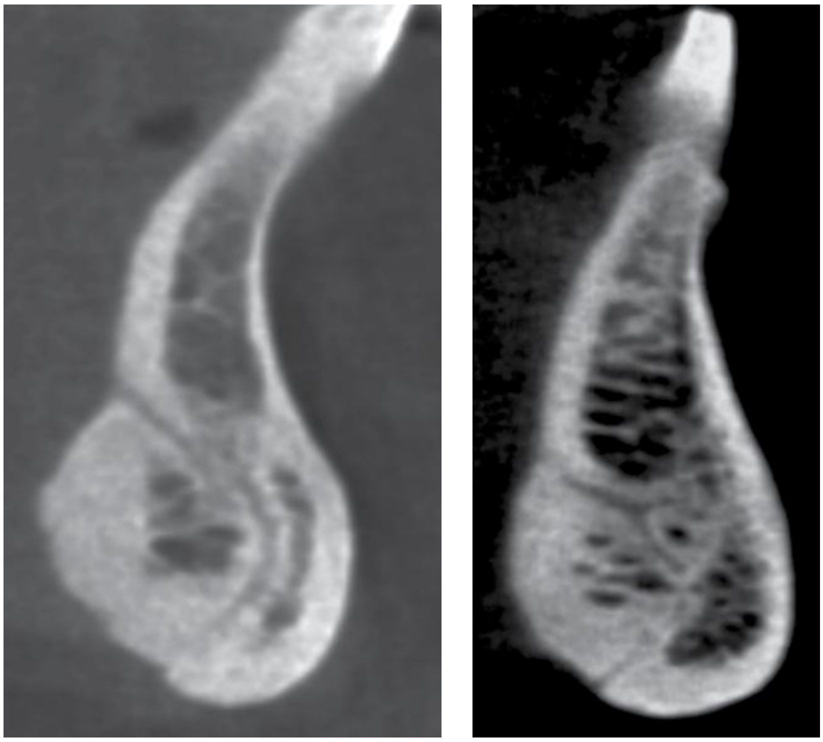


Fig. 6-6 Cone-beam CT cross-sectional images showing the relation and communication between the superior and inferior genial spinal canal (a) and the incisive canal (b).

The observation that immediate loading of implants in the anterior mandible, using the Brånemark Novum® concept rather than a conventional implant-supported overdenture, results in a significant reduction of tactile function could be explained by contact with the aforementioned neurovascular bundles in the anterior mandible (Abarca *et al*. 2006).

Maxillary neuroanatomy

The maxillary nerve is a sensory nerve, with its superior nasal and alveolar branches supplying the palate, nasal and maxillary sinus mucosa, upper teeth and their periodontium. The anterior superior alveolar nerve sometimes runs in a clearly defined canal, palatally of the canine (canalis sinuosus) (Shelley *et al*. 1999). Care should therefore be taken to avoid neurovascular trauma during canine implant installation. Another branch of importance during implant placement is the superior nasal branch, denoted the nasopalatine nerve. It descends to the roof of the mouth through the nasopalatine canal and communicates with the corresponding nerve of the opposite side and with the anterior palatine nerve (Mraiwa *et al*. 2004). Typically, it has been described as forming a Y-shape with the orifices of two lateral canals, and terminating at the nasal floor level in the foramina of Stenson (Fig. 6-7). Occasionally, two additional minor canals carry the nasopalatine nerves (foramina of Scarpa) (Fig. 6-7a). Mraiwa *et al*. (2004) point out a significant variability in both the dimensions and morphologic appearance of the nasopalatine canal. To avoid disturbing neurovascular bundles and causing further complications, implant placement in the maxillary incisor region should consider this important structure (Fig. 6-8).

Histologic background

Periodontal innervation

Periodontal receptors are located within the gingiva, jaw bone, periosteum, and periodontal ligament. Most receptors seem to have mechanoreceptive characteristics, contributing to a sophisticated exteroceptive tactile function. This tactile information is not primarily used for protective purposes, but rather to improve oral motor behavior and fine-tuning of biting and chewing (Trulsson 2006).

It is clear that periodontal ligament plays a predominant role in this dedicated mechanoreceptive function. It has an extremely rich sensory nerve supply, especially in those locations that are more prone to displacement (periapical, buccal, and lingual). It contains three types of nerve endings: free nerve endings, Ruffini-like endings, and lamelated corpuscles (Lambrichts *et al*. 1992). Free nerve endings stem from both unmyelinated and myelinated nerve fibers. Lamellated corpuscles are found in close contact with each other. Most mechanoreceptive endings are however Ruffini-like, and predominantly present in the apical part of the periodontal ligament. Morphologic studies indicate that these endings are in close contact with collagen fibers of the surrounding tissues

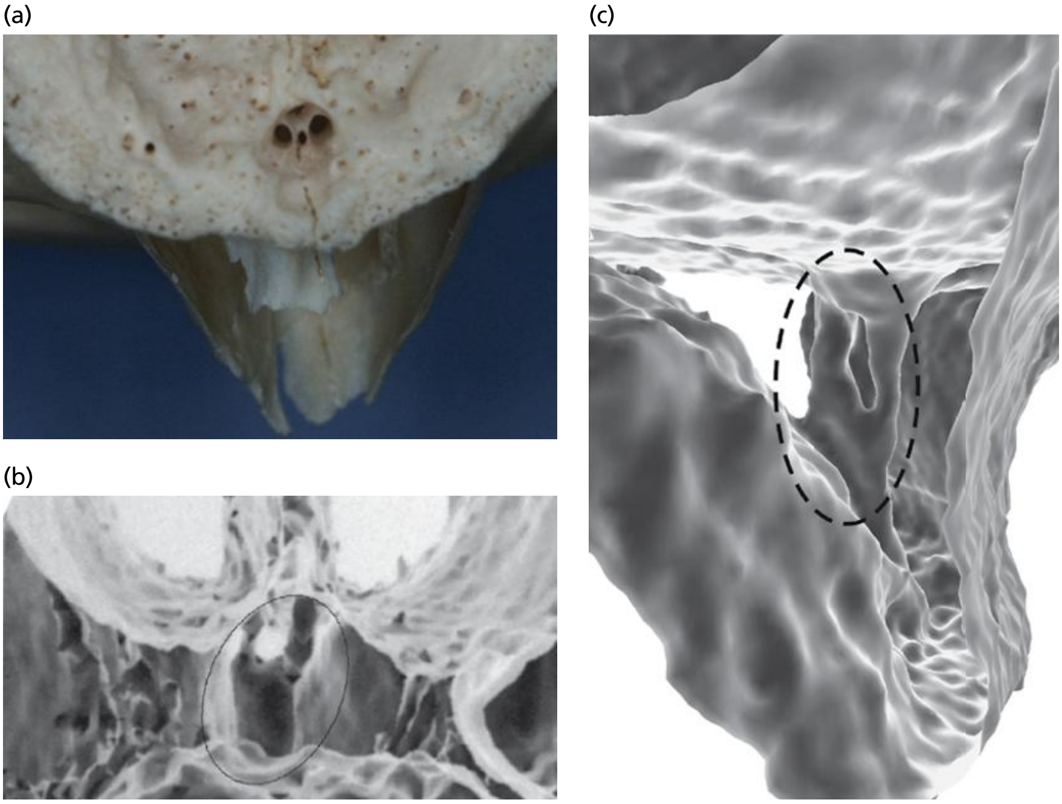


Fig. 6-7 Anatomy of the nasopalatine canal. (a) View from the palate of an edentulous dry skull showing the nasopalatine foramen, formed at the articulation of both maxillae, behind the incisor teeth. In the depth of the canal, the orifices of two lateral canals are seen. As an anatomic variant, two minor canals can be observed on the midline, one anterior and one posterior to the major nasopalatine canals. (b, c) Three-dimensional reconstruction of the palate and the floor of the nose, seen from a posterior angle (b) and from the side view (c). The outline of the common Y morphology of the nasopalatine canal is circled.

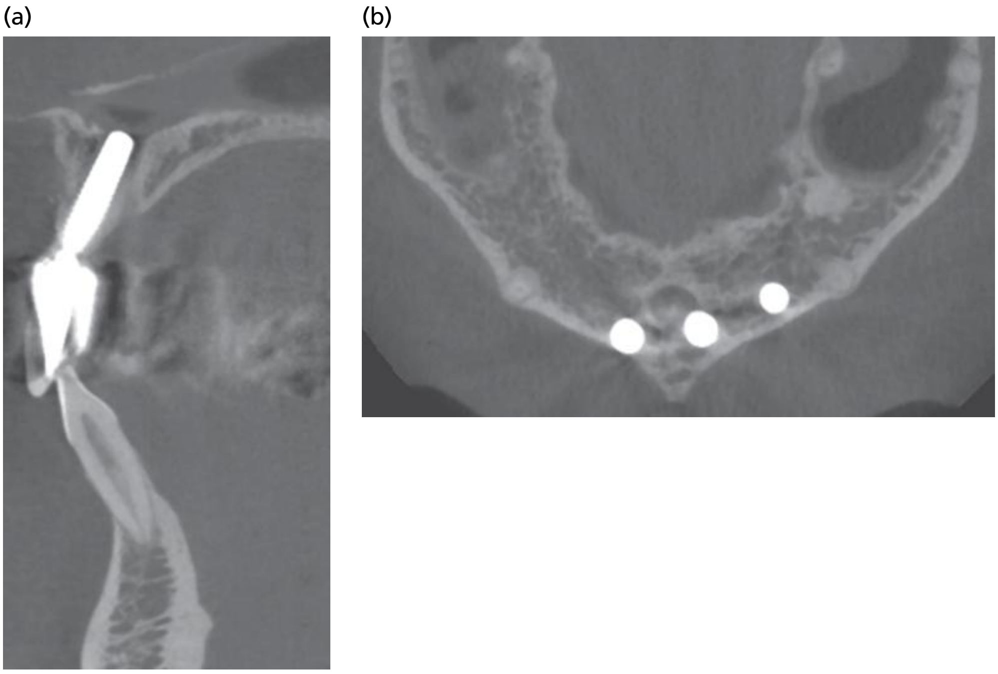


Fig. 6-8 Cone-beam CT sagittal (a) and axial (b) views showing implant 21 in the nasopalatine canal, which prevents osseointegration and creates a pathologic pocket as well as causing sensory disturbances.

(Lambrichts *et al*. 1992) (Fig. 6-9). This particular association may explain their extremely high sensitivity upon loading a tooth. This results in low threshold levels for periodontal tactile function (≤10 μm), and is considered as the basis of an elaborate sensory apparatus that may be linked to a number of clinical phenomena.

The mechanoreceptive function of the periodontal ligament signals the differential information about the mechanical events that occur during biting and

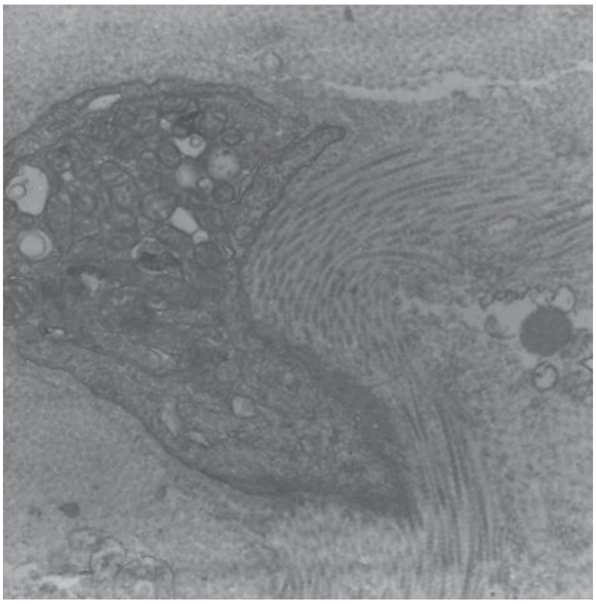


Fig. 6-9 Electron microscope image at the level of the human periodontal ligament, showing collagen fibrils inserted into the basal lamina of an ensheathing cell in a Ruffini-like receptor. (Source: Jacobs & Steenberghe 2006. Reproduced with permission from John Wiley & Sons.)

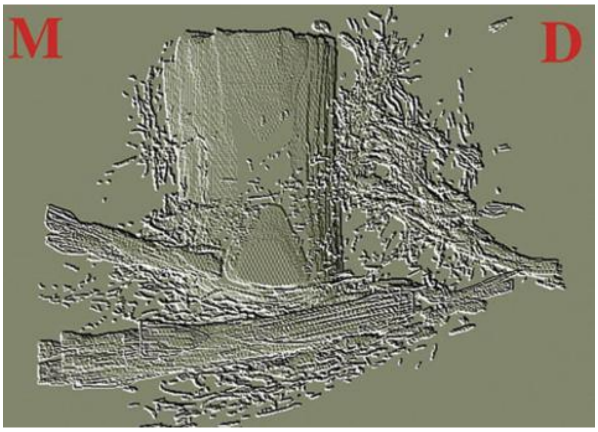


Fig. 6-10 Reconstruction of histologic slices showing the regeneration of nerve tissue 3 months after implantation of a cylindrical oral implant in a dog's jaw bone. (M, mesial; D, distal.) (Source: Wang *et al*. 1998. Reproduced with permission from R. Jacobs, Department of Periodontology, KU Leuven.)

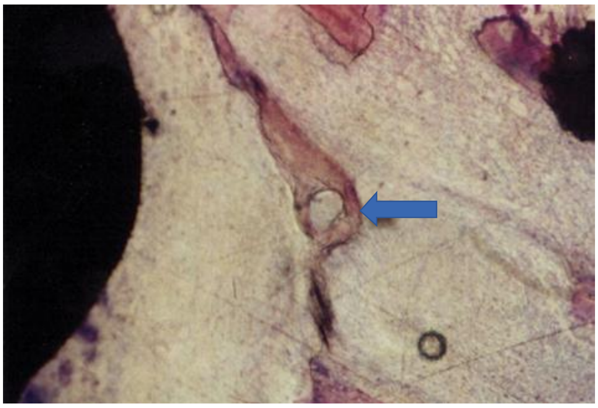


Fig. 6-11 Bone–implant specimen obtained from a cat model and subjected to immunohistochemical detection of neural structures under light microscopy. Elaborate neural structures in the bone trabeculae are seen surrounding the titanium implant. This histologic slice shows the titanium implant–bone tissue, with a bundle of myelinated nerve fibers in the bone trabeculum (arrow). (Source: Lambrichts *et al*. 1992. Reproduced with permission of Ivo Lambrichts, University of Hasselt and R. Jacobs, editor-publisher Osseoperception, Department of Periodontology, KU Leuven).

chewing (Trulsson 2006). The detailed differential signaling allows the brain to analyze and characterize these specific mechanical events, enabling further processing for fine-tuning and resulting in an optimized masticatory sequence (Trulsson 2006). Considering this crucial role, it is clear that some sensory–motor interactions are impaired or even lost when the periodontal ligament is altered or damaged. When teeth are extracted and thus ligament receptors eliminated, tactile functioning may be hampered. Indeed, Haraldson (1983) described a similar muscle activity during the entire masticatory sequence in patients with implant-supported fixed prosthesis. In subjects with natural teeth, the chewing pattern gradually changes as the food bolus properties change. Jacobs and van Steenberghe (1995) identified a silent period in muscle activity (reflex response) when teeth or implants neighboring teeth, but not implants in a fully edentulous jaw bone, were tapped. These findings may illustrate the modulatory role of periodontal ligament input in jaw muscle activity.

Peri-implant innervation

Tooth extraction damages a large number of sensory nerve fibers and corresponds to an amputation, where the target organ and peripheral nervous structures have been destroyed (Mason & Holland 1993). After extraction of teeth, the myelinated fiber content of the inferior alveolar nerve is reduced by 20% (Heasman 1984). This finding indicates that fibers originally innervating the tooth and periodontal ligament are still present in the inferior alveolar nerve. Linden and Scott (1989) succeeded in stimulating nerves of periodontal origin in healed extractions sockets, which implies that some nerve endings remain functional. Nevertheless, most of the surviving mechanoreceptive neurons represented in the mesencephalic nucleus may lose some functionality (Linden & Scott 1989). These experiments have been the basis for a long-lasting debate on the presence and potential function of sensory nerves fibers in the bone and peri-implant environment. Histologic evidence indicates that there may be some re-innervation around osseointegrated implants (Wang *et al*. 1998; Lambrichts 1998) (Figs. 6-10, 6-11). Indeed, it has been shown that the surgical trauma from implantation of endosseous implants may lead to degeneration of surrounding neural fibers. Soon, however, new fibers are observed to sprout and the number of free nerve endings close to the bone–implant interface gradually increases during the first weeks of healing (Wada *et al*. 2001). A more recent study in the dog succeeded in partially regenerating the periodontal ligament on an implant surface (Jahangiri *et al*. 2005). Whether such regeneration can also induce restoration of the peripheral feedback pathway needs further verification.

On the other hand, existing mechanoreceptors in the periosteum may also play a role in tactile function upon implant stimulation. It is evident that oral implants offer a different type of loading and force transfer from teeth, given the intimate bone–implant contact with elastic bone properties instead of the characteristic viscoelasticity of the periodontal ligament. Thus, forces applied to osseointegrated implants are directly transferred to the bone and bone deformation may lead to receptor activation in the peri-implant bone and, more specifically, in the neighboring periosteum.

Testing tactile function

Exteroceptive function can be examined by neurophysiologic as well as psychophysical methods.

Neurophysiologic assessment

Neurophysiologic investigations of the sensory function of the human trigeminal system are risky and therefore rarely reported (Johansson *et al*. 1988a, b; Trulsson *et al*. 1992). Alternatively, non-invasive approaches may be considered to evaluate oral tactile function. The first approach is the recording of the so-called trigeminal somatosensory evoked potentials (TSEPs) after stimulation of receptors in the oral cavity. Unfortunately, SEPs from the trigeminal branches are, in contrast to those recorded from limbs, weak and difficult to discriminate from the background noise. Advanced signal analysis is required to gain reliable information (van Loven *et al*. 2000, 2001). Another non-invasive method to assess sensory function is to visualize brain activities by functional magnetic resonance imaging (fMRI) (Borsook *et al*. 2006). This is a complex method, which has received hardly any attention in relation to tactile function of teeth and implants (Lundborg *et al*. 2006; Miyamoto *et al*. 2006; Habre-Hallage *et al*. 2010). When combining fMRI with other techniques such as psychophysics and TSEPs, it may offer a new non-invasive approach to evaluate human oral somatosensory function.

Psychophysical assessment

Sensory function can also be evaluated by psychophysical testing, relying on the patient's response. This technique has often been applied to test oral tactile function (Jacobs *et al*. 2002b–d). A major advantage of psychophysical testing is that it employs simple non-invasive techniques that may be performed in a clinical environment. When carried out under strictly standardized conditions, the psychophysical response can be directly linked to the neural receptor activation (Vallbo & Johansson 1984).

Periodontal tactile function: Influence of dental status

Oral tactile function is influenced by tooth position and dental status (Jacobs & van Steenberghe 2006). The tactile function of teeth is primarily determined by the presence of periodontal ligament receptors. Vital or non-vital teeth may show a comparable tactile function. However, when periodontal ligament receptors are reduced or eliminated (e.g. through periodontitis, bruxism, chewing, extraction, anesthesia), tactile function is impaired (Table 6-1). Clinically, this implies that a patient's ability to detect occlusal inaccuracies (e.g. induced by restorative treatment) is decreased in these situations. Indeed, exteroceptors inform the nervous system on the characteristics of the stimulus, which then allows modulation of the motor neuron pool to optimize jaw motor activity and avoid overloading. Elimination of these exteroceptors by tooth extraction may reduce the tactile function to an important extent (for review see Jacobs & van Steenberghe 2006). Even after rehabilitation with a prosthesis, tactile function remains impaired and inappropriate exteroceptive feedback may thus present a risk for overloading the prosthesis (Jacobs & van Steenberghe 2006). In comparison to the tactile function of natural dentitions, the active thickness detection threshold is seven to eight times higher for dentures, but only three to five times higher for implants (see Table 6-1). For the passive detection of forces applied to upper teeth, thresholds for dentures are increased 55 times and for implants 50 times (see Table 6-1). The large discrepancies between active and passive thresholds can be explained by the fact that several receptor groups may respond to active thickness detection, while the passive force

Table 6-1 Factors influencing the tactile function of teeth (for review see Jacobs *et al*. 2002b; Jacobs & van Steenberghe 2006)

Influencing factor	Active threshold: thickness detection	Passive threshold: force detection
Vital tooth	20 μm	2 g
Avital tooth	20 μm	2 g
Anesthesia	↑	↑
Periodontitis	↑	↑ (>5 g)
Chewing	↑	↑
Bruxism	↑	↑
Extraction	↑	↑
Re-implantation	↑	↑
Denture	150 μm	150 g
Implant-supported prosthesis	50 μm	100 g
Aging	↑	↑
Polyneuropathy	↑	↑

↑, increase in threshold level implies decrease in tactile function and hampered feedback.

application selectively activates only periodontal ligament receptors. The latter are eliminated after extraction, which may explain the reduced tactile function in edentulous patients.

After rehabilitation with a bone-anchored prosthesis however, edentulous patients seem to function quite well. These patients perceive mechanical stimuli exerted on osseointegrated implants in the jaw bone. Some even note a special sensory awareness with the bone-anchored prosthesis, coined "osseoperception". This can be defined as perception of external stimuli transmitted via the implant through the bone by activation of receptors located in the peri-implant environment, periosteum, skin, muscles, and/or joints, with periosteal innervation playing a key role (Jacobs 1998). From the existence of the osseoperception phenomenon it can be hypothesized that the feedback pathway to the sensory cortex is partly restored with a representation of the prosthesis in the sensory cortex, which modulates of the motor neuron pool, leading to a more natural functioning and avoidance of overload.

From periodontal tactile function to peri-implant osseoperception

Neurophysiologic evidence for the cortical plasticity with representation of the implant in the sensory cortex can be found in some experiments evoking TSEPs upon implant stimulation. By triggering sweeps in the electroencephalogram by means of an implant stimulation device and by cumulative and advanced analysis of the sweeps, significant waves can be noted (Fig. 6-12). These experiments indicate that endosseous and/or periosteal receptors around the implants convey the sensation (Van Loven *et al*. 2000). These mechanisms could be the basis for implant-mediated sensory–motor control, which may have important clinical implications for a more natural functioning with implant-supported prostheses.

Evidence for cortical plasticity may open the way to complete integration of implants in the human body. Henry *et al*. (2005) extracted lower incisors in mole rats and with fMRI analysis showed a reorganization of the orofacial representation in the *primary sensory cortex* 5–8 months later. This study may indicate that a cortical representation of teeth may lead to significant restructuring after tooth loss. Likewise, an fMRI study by Lundborg *et al*. (2006) demonstrated that upon tactile stimulation of an osseointegrated prosthetic thumb, the primary somatosensory cortex is bilaterally activated in an area corresponding to that of the hand. As only stimulation of the healthy thumb would be expected to activate the contralateral cortex, the presence of bilateral cortical activation upon implant stimulation may be explained by some compensatory mechanism that recruits additional sensory areas after amputation (Lundborg *et al*. 2006). Following the stimulation protocol of Habre-Hallage *et al*. (2010) with 1-Hz punctuate tactile stimuli applied to teeth and implants, this group recently found that upon stimulation of implants and teeth, the somatosensory cortex was activated (Habre-Hallage 2011). The implant activated a larger bilateral cortical network outside the somatosensory areas. This novel study demonstrates that punctuate mechanical stimulation of oral implant activates cortical somatosensory areas and induces brain plasticity (Habre-Hallage 2011) (Fig. 6-13). This activation may represent the underlying mechanism of osseoperception and confirms that some related cortical plasticity may truly exist.

However, the central neural pathways and neural characteristics contributing to implant-mediated sensory–motor control remain unclear. Future research should therefore try to visualize cortical plasticity after tooth extraction and further functional rehabilitation with implants in humans. It should be considered that an extraction and immediate implant rehabilitation protocol might induce a different cortical remodeling from that with a traditional two-stage implant rehabilitation protocol. An interesting phenomenon with respect to sensory–motor integration of osseointegrated implants may be the so-called phantom tooth (after extraction) or phantom limb (after amputation), allowing perception of lost body parts (Jacobs *et al*.

(a) (b)

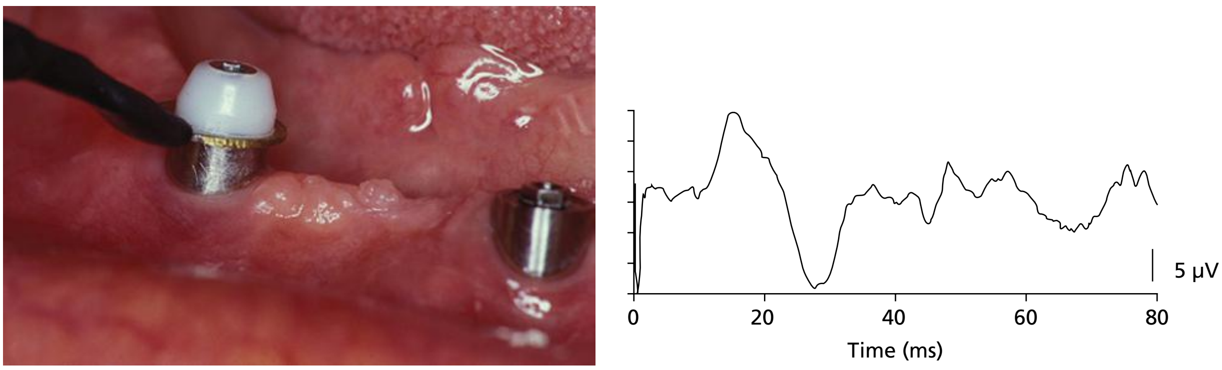


Fig. 6-12 (a) Electrical stimulation of an osseointegrated implant using a ring-shaped stimulation electrode fixed by a cover screw. (b) Trigeminal evoked potential elicited by electrical stimulation of an osseointegrated implant in the mandible. A similar potential could be maintained after topical anesthesia of the peri-implant soft tissues, indicating that the trigeminal potentials originated from other peri-implant structures such as bone and periosteal receptors.

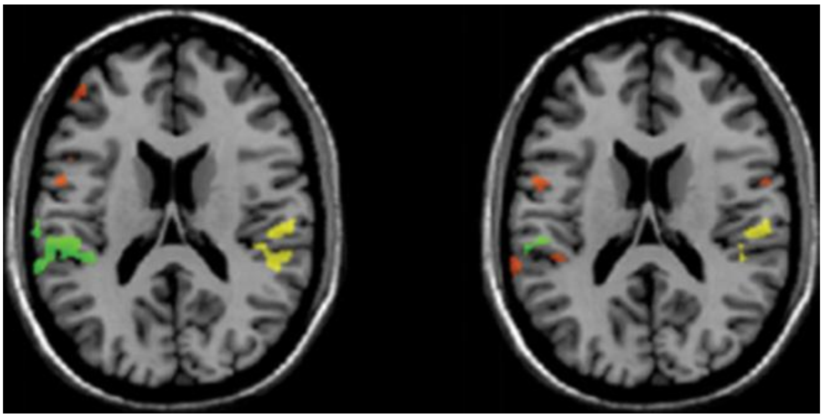


Fig. 6-13 Functional MRI showing cortical activation during tactile stimulation (1 Hz) of implant 21 (a) and tooth 23 (b) in the same subject (random effect analysis of subject sample). The primary somatosensory area is displayed in green, the secondary somatosensory area in yellow, and the other foci of activation in orange. (Source: Habre-Hallage *et al*. 2012, with permission of Pascale Habre, KU Leuven.)

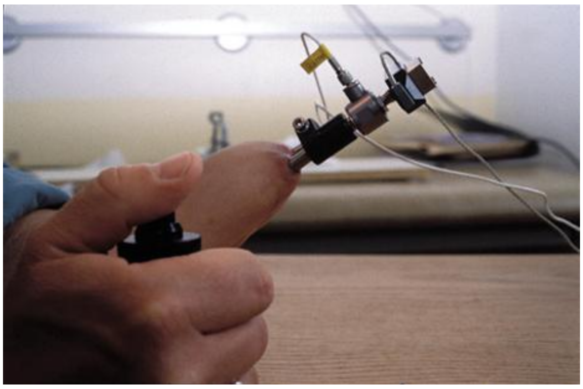
2002c). In fact, it could be assumed that such a phantom feeling of the lost limb may overlap with or enforce the feeling of a bone-anchored prosthetic limb (Jacobs *et al*. 2000). In this way, phantom sensations might contribute to physiologic integration of a bone-anchored prosthesis in the human body.

Once neuroplasticity after amputation and osseointegration is fully unraveled, it may be a consideration during treatment as it could optimize adaptation to oral rehabilitation and implant placement (Feine *et al*. 2006).

From osseoperception to implant-mediated sensory–motor interactions

During the last few decades, millions of patients have been rehabilitated by means of osseointegrated implants. Even though part of the peripheral feedback mechanism is lost after tooth extraction, edentulous patients seem to function quite well, especially when rehabilitated with a prosthesis retained by or anchored to osseointegrated implants (Jacobs 1998). These findings correspond well to those in amputees rehabilitated with a bone-anchored prosthesis rather than a socket prosthesis. During skeletal reconstruction, psychophysical testing reveals an improved tactile and vibrotactile capacity with osseointegrated implants and bone-anchored prosthetic limbs (Fig. 6-14). Furthermore, both edentulous patients and amputees seem to report an improved awareness and special feeling with the implant-supported prosthesis, suggesting the hypothesis that the peripheral feedback pathway is partially restored with a representation of the artificial limb feeling in the sensory cortex (Lundborg *et al*. 2006). If this hypothesis can be confirmed, osseointegrated implants in the jaw or other skeletal bones might contribute to implant-mediated sensory motor control, allowing physiologic integration of the implant into the human body and as a result, a more natural functioning (Jacobs & van Steenberghe 2006).

(a)



(b)

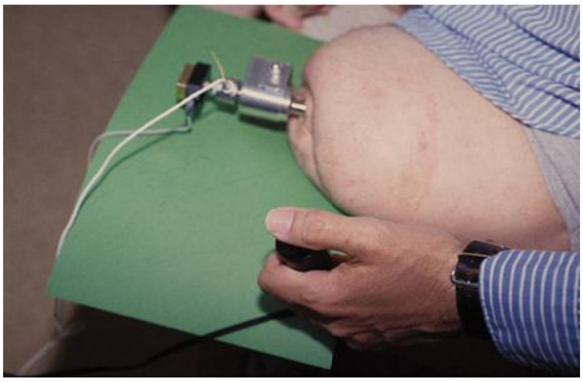


Fig. 6-14 Psychophysical test set-up using patient-controlled remote control for a vibrotactile stimulator fixed to a radial (a) and femoral (b) osseointegrated implant. This particular test set-up yields a superior perception for implants and bone-anchored prosthetic limbs as compared to socket prostheses. (Source: Jacobs *et al*. 2000. Reproduced with permission from Sage.)

Clinical implications of implant-mediated sensory–motor interactions

Psychophysical testing of various bone-anchored prostheses confirms an improved tactile function leading to a better physiologic integration of the limb. If perception upon implant stimulation is working well, peripheral feedback mechanism may be restored and help fine-tune motor control. This implant-mediated sensory–motor interaction may thus help to achieve a more natural function with the bone-anchored prosthesis. Osseointegrated thumb prostheses even allow patients to perform the activities of daily life without any problem, which can be attributed to bone-anchorage and bilateral cortical representation after prosthesis stimulation. Considering the increased tactile threshold level for oral implant stimulation, a few clinical implications should be considered. During rehabilitation by means of implant-supported prostheses, a dentist should not rely on the patient's perception of occlusion. In this respect, the dentist should also be aware of the gradually increasing tactile function during the healing period after implant placement. This may be of particular importance when dealing with immediate loading protocols. To avoid any overloading related to suboptimal feedback mechanisms, patients should be encouraged to limit chewing forces by eating only soft foods during the healing period. Furthermore, parafunctional habits such as grinding or clenching might have a negative impact during the implant healing phase, but further research is needed to confirm this assumption (Lobbezoo *et al*. 2006). Until further evidence is available, bruxism may be considered as a relative contraindication to immediate loading protocols (Glauser *et al*. 2001).

Conclusion

Sensory feedback plays an essential role in the fine-tuning of jaw and limb motor control. Periodontal mechanoreceptors, and more specifically those located in the periodontal ligament, are extremely sensitive to external mechanical stimuli. These receptors play the key role in tactile function of teeth, yielding detection thresholds of a thickness of about 20 μm between antagonistic teeth and 1–2 g upon tooth loading. It is the sensory characteristics and related peripheral feedback that dedicate the periodontal ligament receptors to the fine-tuning of masticatory and other oral motor behaviors.

It is clear that any condition that may influence periodontal mechanoreceptors may also alter the sensory feedback pathway, and thus influence tactile function and modulation of jaw motor control (e.g. periodontal breakdown, bruxism, re-implantation, anesthesia). After extraction of teeth, the periodontal ligament is lost and so are its mechanoreceptors. After placement of oral implants, detection thresholds are increased to a thickness of at least 50-100 μm and 50–100 g upon tooth loading.

Surprisingly, patients rehabilitated by means of osseointegrated implants seem to function quite well and/or sense even better. In accordance with this, amputated patients rehabilitated with a lower limb prosthesis anchored to the bone by means of an osseointegrated implant have reported that they could sense the type of surface they were walking on, and those with a bone-anchored thumb prosthesis had a conscious perception of fingers.

The underlying mechanism of this so-called "osseoperception" phenomenon remains a matter of debate, but it is assumed that mechanoreceptors in the peri-implant bone and neighboring periosteum may be activated upon implant loading. Histologic, neurophysiologic, and psychophysical evidence of osseoperception has been amassed, making the assumption more likely that a proper peripheral feedback pathway can be restored when using osseointegrated implants. This implant-mediated sensory–motor control may have important clinical implications, because a more natural functioning can be attempted with implant-supported prostheses. This may open doors for physiologic and psychophysical integration of implants in the human body.

References

Abarca, M., van Steenberghe, D., Malevez, C., De Ridder, J. & Jacobs, R. (2006). Neurosensory disturbances after immediate loading of implants in the anterior mandible: an initial questionnaire approach followed by a psychophysical assessment. *Clinical Oral Investigations* **10**, 269–277.

Borsook, D., Burstein, R., Moulton, E. & Becerra, L. (2006). Functional imaging of the trigeminal system: applications to migraine pathophysiology. *Headache* **46** *Suppl 1*, S32–S38.

Feine, J., Jacobs, R., Lobbezoo, F. *et al*. (2006). A functional perspective on oral implants – state-of-the-science and future recommendations. *Journal of Oral Rehabilitation* **33**, 309–312.

Glauser, R., Ree, A., Lundgren, A. *et al.* (2001). Immediate occlusal loading of Brånemark implants applied in various jawbone regions: a prospective, 1-year clinical study. *Clinical Implant Dentistry and Related Research* **3**, 204–213.

Habre-Hallage, P. (2011). Assessment of changes in oral tactile function and osseoperception by oral endosseous implant placement. PhD Thesis. Leuven: KU Leuven.

Habre-Hallage, P., Hermoye, L., Gradkowski, W. *et al*. (2010). A manually controlled new device for punctuate mechanical stimulation of teeth during functional magnetic resonance imaging studies. *Journal of Clinical Periodontology* **37**, 863–872.

Haraldson, T. (1983). Comparisons of chewing patterns in patients with bridges supported on osseointegrated implants and subjects with natural dentitions. *Acta Odontologica Scandinavica* **41**, 203–208.

Heasman, P.A. (1984). The myelinated fibre content of human inferior alveolar nerves from dentate and edentulous subjects. *Journal of Dentistry* **12**, 283–286.

Henry, E.C., Marasco, P.D. & Catania, K.C. (2005). Plasticity of the cortical dentition representation after tooth extraction in naked mole-rats. *Journal of Comparative Neurology* **485**, 64–74.

Jacobs, R. (1998). *Osseoperception*. Leuven: Department of Periodontology, Catholic University Leuven.

Jacobs, R. & van Steenberghe, D. (1994). Role of periodontal ligament receptors in the tactile function of teeth: a review. *Journal of Periodontal Research* **29**, 153–167.

Jacobs, R. & van Steenberghe, D. (1995). Qualitative evaluation of the masseteric poststimulus EMG complex following mechanical or acoustic stimulation of osseointegrated oral implants. *International Journal of Oral & Maxillofacial Implants* **10**, 175–182.

Jacobs, R. & van Steenberghe, D. (2006). From osseoperception to implant-mediated sensory-motor interactions and related clinical implications. *Journal of Oral Rehabilitation* **33**, 282–292.

Jacobs, R., Brånemark, R., Olmarker, K. *et al.* (2000). Evaluation of the psychophysical detection threshold level for vibrotactile and pressure stimulation of prosthetic limbs using bone anchorage or soft tissue support. *Prosthetics and Orthotics International* **24**, 133–142.

Jacobs, R., Mraiwa, N., van Steenberghe, D, Gijbels, F. & Quirynen, M. (2002a). Appearance, location, course, and morphology of the mandibular incisive canal: an assessment on spiral CT scan. *Dentomaxillofacial Radiology* **31**, 322–327.

Jacobs, R., Wu, C-H., Goossens, K. *et al.* (2002b). Oral versus cutaneous sensory testing: a review of the literature. *Journal of Oral Rehabilitation* **29**, 923–950.

Jacobs, R., Wu, C-H., Goossens, K. *et al*. (2002c). A case-control study on the psychophysical and psychological characteristics of the phantom tooth phenomenon. *Clinical Oral Investigations* **6**, 58–64.

Jacobs, R., Wu, C-H., Desnyder, M., Kolenaar, B. & van Steenberghe, D. (2002d). Methodologies of oral sensory tests. *Journal of Oral Rehabilitation* **29**, 720–730.

Jacobs, R., Mraiwa, N., van Steenberghe, D., Sanderink, G. & Quirynen, M. (2004). Appearance of the mandibular incisive canal on panoramic radiographs. *Surgical and Radiologic Anatomy* **26**, 329–333.

Jacobs, R., Lambrichts, I., Liang, X. *et al*. (2007). Neurovascularisation of the anterior jaw bones revisited using high resolution magnetic resonance imaging. *Oral Surgery, Oral Medicine, Oral Pathology, Oral Radiology and Endodontics* **103**, 683–693.

Jahangiri, L., Hessamfar, R. & Ricci, J.L. (2005). Partial generation of periodontal ligament on endosseous dental implants in dogs. *Clinical Oral Implant Research* **16**, 396–401.

Johansson, R.S., Trulsson, M., Olsson, K.Å. & Westberg, K.-G. (1988a). Mechanoreceptor activity from the human face and oral mucosa. *Experimental Brain Research* **72**, 204–208.

Johansson, R.S., Trullson, M., Olsson, K.Å. & Abbs, J.H. (1988b). Mechanoreceptive afferent activity in the infraorbital nerve in man during speech and chewing movements. *Experimental Brain Research* **72**, 209–214.

Lambrichts, I. (1998). Histological and ultrastructural aspects of bone innervation. In: Jacobs, R., ed. *Osseoperception*. Leuven: Department of Periodontology, KU Leuven.

Lambrichts, I., Creemers, J. & van Steenberghe, D. (1992). Morphology of neural endings in the human periodontal ligament: an electronmicroscopic study. *Journal of Periodontal Research* **27**, 191–196.

Liang, X., Jacobs, R., Lambrichts, I. *et al*. (2005a). Microanatomical and histological assessment of the content of superior genial spinal foramen and its bony canal. *Dentomaxillofacial Radiology* **34**, 362–368.

Liang, X, Jacobs, R. & Lambrichts, I. (2005b). Appearance, location, course and morphology of the superior and inferior genial spinal foramina and their bony canals: an assessment on spiral CT scan. *Surgical and Radiologic Anatomy* **9**, 1–7.

Linden, R.W. & Scott, B.J. (1989). The effect of tooth extraction on periodontal ligament mechanoreceptors represented in the mesencephalic nucleus of the cat. *Archives of Oral Biology* **34**, 937–941.

Lobbezoo, F., Van Der Zaag, J. & Naeije, M. (2006). Bruxism: its multiple causes and its effects on dental implants – an updated review. *Journal of Oral Rehabilitation* **33**, 293–300.

Lundborg, G., Waites, A., Björkman, A., Rosén, B. & Larsson, E.-M. (2006). Functional magnetic resonance imaging shows cortical activation on sensory stimulation of an osseointegrated prosthetic thumb. *Scandinavian Journal of Plastic and Reconstructive Surgery and Hand Surgery* **40**, 234–239.

Martin, J.H. (1991). Coding and processing of sensory information. In: Kandel, E.R., Schwartz, J.H. & Jessel, T.M., eds. *Principles of Neural Science*, 3rd edn. Norwalk, CT: Appleton & Lange, pp. 329–340.

Mason, A.G. & Holland, G.R. (1993). The reinnervation of healing extraction sockets in the ferret. *Journal of Dental Research* **72**, 1215–1221.

Miyamoto, J.J., Honda, M., Saito, D.N. *et al*. (2006). The representation of the human oral area in the somatosensory cortex: a functional MRI study. *Cerebral Cortex* **16**, 669–675.

Mraiwa, N., Jacobs, R., Moerman, P. *et al*. (2003a). Presence and course of the incisive canal in the human mandibular interforaminal region: two-dimensional imaging versus anatomical observations. *Surgical and Radiologic Anatomy* **25**, 416–423.

Mraiwa, N., Jacobs, R., van Steenberghe, D. & Quirynen, M. (2003b). Clinical assessment and surgical implications of anatomic challenges in the anterior mandible. *Clinical Implant Dentistry and Related Research* **5**, 219–225.

Mraiwa, N., Jacobs, R., Van Cleynenbreugel J.*et al*. (2004). The nasopalatine canal revisited using 2D and 3D CT imaging. *Dentomaxillofacial Radiology* **33**, 396–402.

Oliveira-Santos, C., Souza, P.H., De Azambuja Berti-Couto, S. *et al*. (2011). Characterisation of additional mental foramina through cone beam computed tomography. *Journal of Oral Rehabilitation* **38**, 595–600.

Penfield, W. & Rasmussen, T. (1950). *The Cerebral Cortex of Man: A Clinical Study of Localization of Function.* New York: Macmillan.

Shelley, A.M., Rushton, V.E. & Horner, K. (1999). Canalis sinuosus mimicking a periapical inflammatory lesion. *British Dental Journal* **186**, 378–379.

Trulsson, M. (2006). Sensory-motor function of periodontal mechanoreceptors. *Journal of Oral Rehabilitation* **33**, 262–273.

Trulsson, M., Johansson, R.S. & Olsson, K.Å. (1992). Directional sensitivity of human periodontal mechanoreceptive afferents to forces applied to the teeth. *Journal of Physiology (London)* **447**, 373–389.

Vallbo, Å.B. & Johansson, R.S. (1984). Properties of cutaneous mechanoreceptors in the human hand related to touch sensation. *Human Neurobiology* **3**, 3–14.

Van Loven, K., Jacobs, R., Swinnen, A. *et al.* (2000). Sensations and trigeminal somatosensory-evoked potentials elicited by electrical stimulation of endosseous oral implants in humans. *Archives Oral Biology* **45**, 1083–1090.

Van Loven K, Jacobs, R., Van Hees, J., Van Huffel, S. & van Steenberghe, D. (2001). Trigeminal somatosensory evoked potentials in humans. *Electromyography and Clinical Neurophysiology* **41**, 357–375.

Wada, S., Kojo, T., Wang, Y.H. *et al.* (2001). Effect of loading on the development of nerve fibers around oral implants in the dog mandible. *Clinical Oral Implants Research* **12**, 219–224.

Wang, Y.-H., Kojo, T., Ando, H. *et al*. (1998). Nerve regeneration after implantation in peri-implant area: a histological study on different implant materials in dogs. In: Jacobs, R., ed. *Osseoperception*. Leuven: Department of Periodontology, Catholic University Leuven, pp. 3–11.

Part 2: Epidemiology